0046洛谷P5735 距离函数

```
#include<iostream>
#include<cmath>
#include<iomanip>
using namespace std;

double dis(double x1, double y1, double x2, double y2) {
```

在自定义函数引入因变量时，每个变量要独立声明其类型;如double x1,double x2，不能直接简写成double x1,x2;

```
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
```

制作一个读取x1,y1,x2,y2就可以算出两点之间距离的函数工具

```
int main() {
    double x1, y1, x2, y2, x3, y3;
    cin >> x1 >> y1;
    cin >> x2 >> y2;
    cin >> x3 >> y3;
```

依次读取各个坐标数据

```
    double L1 = dis(x1, y1, x2, y2);
    double L2 = dis(x1, y1, x3, y3);
    double L3 = dis(x2, y2, x3, y3);
```

应用我们建立的距离公式，算出L1，L2，L3;

```
    double ans = L1 + L2 + L3;
    cout << fixed << setprecision(2) << ans << endl;
    return 0;
}
```

完整代码

```
#include<iostream>
#include<cmath>
#include<iomanip>
using namespace std;

double dis(double x1, double y1, double x2, double y2) {

    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));

int main() {
    double x1, y1, x2, y2, x3, y3;
    cin >> x1 >> y1;
    cin >> x2 >> y2;
    cin >> x3 >> y3;

    double L1 = dis(x1, y1, x2, y2);
    double L2 = dis(x1, y1, x3, y3);
    double L3 = dis(x2, y2, x3, y3);

    double ans = L1 + L2 + L3;
```

```
    cout << fixed << setprecision(2) << ans << endl;
    return 0;
}
```